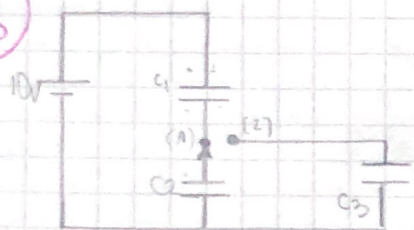


5



$$C_1 = 12 \mu F, C_2 = 6 \mu F, C_3 = 2 \mu F$$

a) Q_{C_3}

b) $V_{C_1 C_2} / V_{C_2 C_3}$

a) C_1 y $C_2 \Rightarrow$ en serie con sw(1) cerrado

$$C_{eq_{C_1 C_2}} = \frac{1}{\frac{1}{12} + \frac{1}{6}} = 4 \mu F$$

$$Q_{eq} = Q_1 = Q_2$$

$$Q_{eq} = C_{eq} \cdot V$$

$$Q_{eq} = 4 \mu F \cdot 10 V$$

$$Q_{eq} = 40 \mu C$$

Con sw 1 abierto y sw (2) cerrado C_2 y $C_3 \Rightarrow$ en paralelo

$$C_{eq} = C_2 + C_3 = 6 + 2 = 8 \mu F$$

$$Q_3 = 2 \mu F \cdot 5$$

$$V = \frac{Q_T}{C_{eq}} = \frac{40 \mu C}{8 \mu F} = 5 V$$

$$Q_3 = 10 \mu C$$

NOTA

$$D- U = \frac{1}{2} \frac{Q^2}{C_{eq_{1,2}}}$$

$$U = \frac{1}{2} \frac{40^2}{4}$$

$$U_{4,2} = 200 \mu J$$

$$U_{C_1, C_2, C_3} = U_1 + U_{2,3}$$

$$U_{C_1, C_2, C_3} = \frac{1}{2} \frac{Q_1^2}{C_1} + \frac{1}{2} \frac{Q_{2,3}^2}{C_{eq_{2,3}}}$$

$$U_{C_1, C_2, C_3} = \frac{1}{2} \frac{(40)^2}{12} + \frac{1}{2} \frac{(40)^2}{8}$$

$$U_{C_1, C_2, C_3} = \frac{200}{3} + 100$$

$$U_{C_1, C_2, C_3} = 166,6 \mu J$$